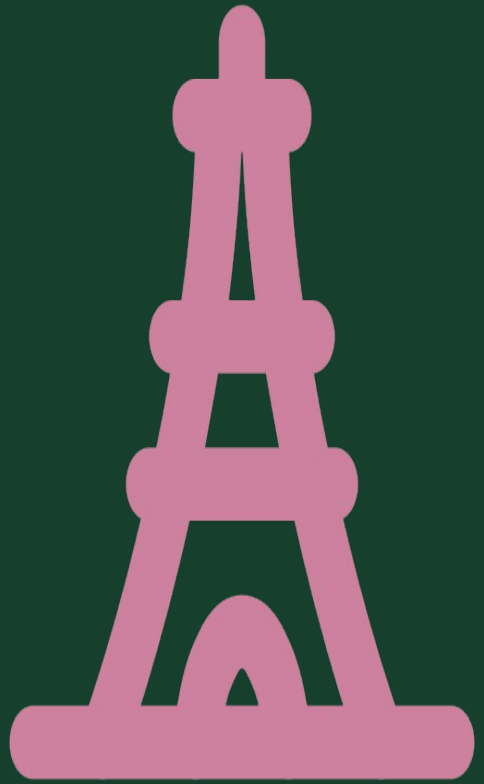
Bonjour, GrandPied !

# SpamJam

Catching spam without losing a client email



Victoria Alajemba & Sheila Kana · ML × Python final project · Souheil Hanoune · November 2026

Le problème

# Blocking a client email is the costly mistake



## False positive

*Costly*

A client email lands in the spam folder: a missed inquiry, maybe a lost contract.



## False negative

*Acceptable*

A spam reaches the inbox: annoying, but staff can delete it.

So we choose models by **average precision**, then set the decision threshold to keep precision on spam high.

Les données

# Spambase: 4,210 unique emails, 57 numeric features

**4,601**

emails in Spambase (UCI, 1999)

**391**

exact duplicates removed

**40 %**

spam, kept in both sets by a stratified split

**3,368 / 842**

training / test emails (80/20, seed 42)

**Watch out:** all legitimate emails come from one person (George, at HP Labs). Words like george, hp or 650 describe him, not spam.

Also: capital-letter counts are very skewed (so we log them), and 857 and 415 are near-duplicates.

La méthode

# One pipeline, tuned on the training set only




*once, before training*

*inside the scikit-learn Pipeline, learned per CV fold*

## Validation

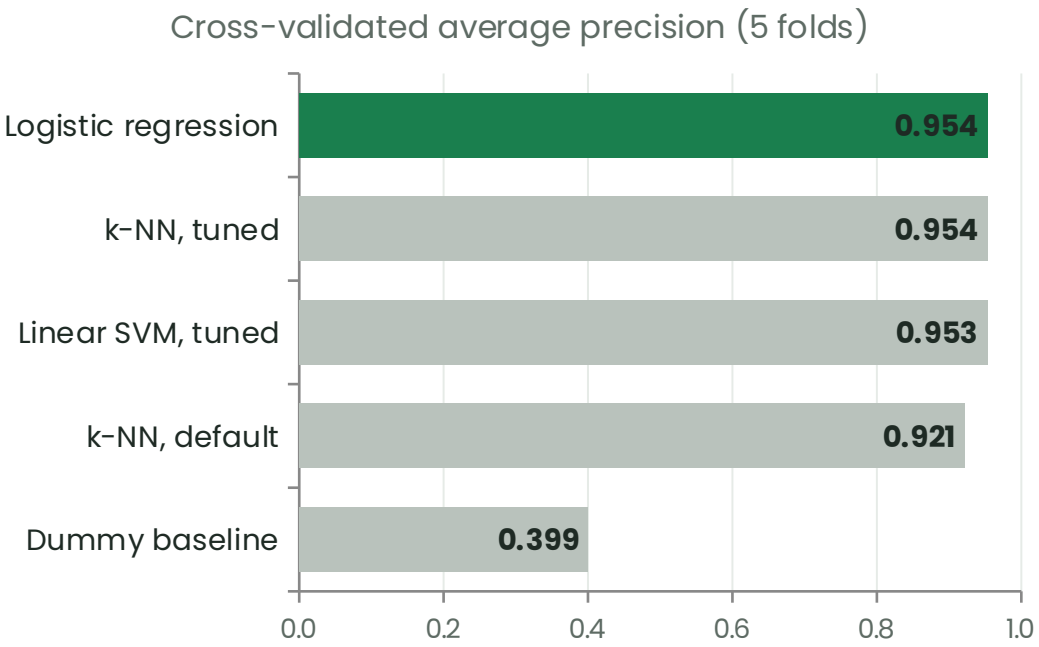
- Stratified 5-fold CV + GridSearchCV, scored by average precision
- A model wins only by > 1 standard error
- Threshold from out-of-fold predictions
- Test set used once, at the very end

## Models compared

- Dummy baseline (always ham)
- Logistic regression (L1/L2, C)
- k-NN (neighbours, weighting)
- Linear SVM (C)

Les résultats

# Three models tied: we keep the simplest




## Why logistic regression

- All three within 0.001: far below one standard error (0.005)
- Simplest: probabilities and readable weights

## Along the way

- Log on capitals: +0.011
- Tuning k-NN: 0.921 → 0.954

Le test final

# On unseen emails, 95 % of flagged emails were spam

**0.782**
decision threshold, from out-of-fold predictions

**0.950**
precision: flagged emails that are spam

**0.795**
recall: about 4 spam out of 5 caught

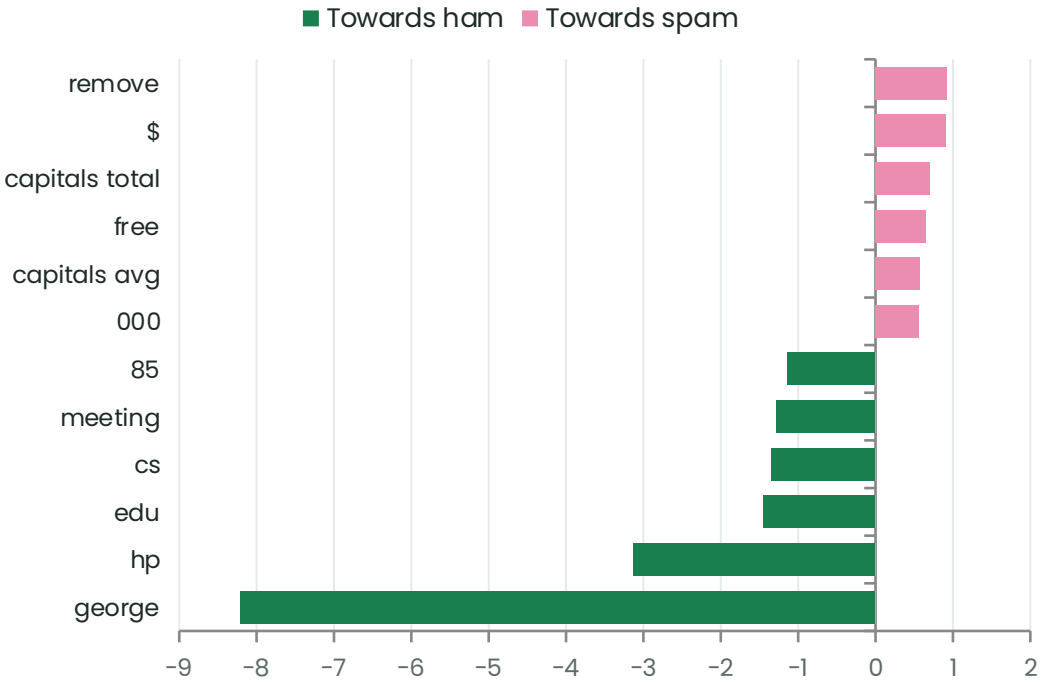
**14 / 506**
legitimate emails flagged

| | predicted ham | predicted spam |
|---|---|---|
| actual ham | 492 | 14 |
| actual spam | 69 | 267 |

**The test set was used once, at the end.** Precision matches the 0.95 we aimed for (margin 0.92 to 0.97). Without the donor's words: same precision (0.952), recall 0.71 instead of 0.79. Flagged emails go to quarantine, not the bin.

Ce que le modèle a appris

# The donor's name is the model's strongest signal




## The donor effect

Without the 9 donor words, average precision drops from 0.954 to 0.938 (about 3 standard errors).

That gain only exists in George's mailbox.

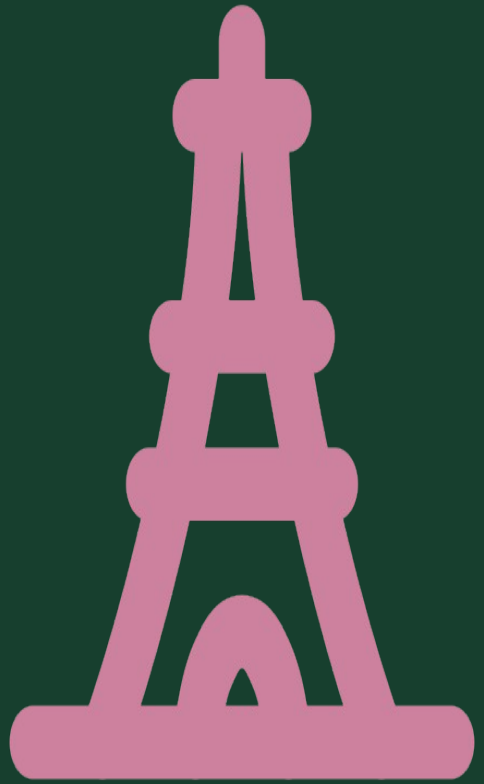
Trained without them, the model isn't fooled by "Hi George," (0 % → 100 % spam).

**GrandPied must retrain on its own emails.**

Merci !

# Pour GrandPied

- Use: a quarantine assistant that catches 4 spam in 5; staff rescue 14 real emails in 506
- Not as it is: English only, flags marketing emails, learned one person's mailbox
- **Next: retrain on GrandPied's own sorted emails, per team**



**Demo: pytest · spamjam predict  Questions ?**